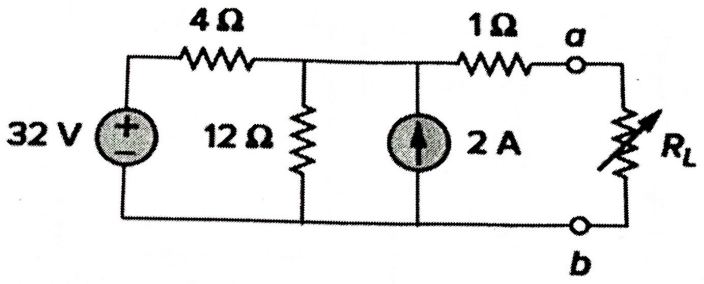
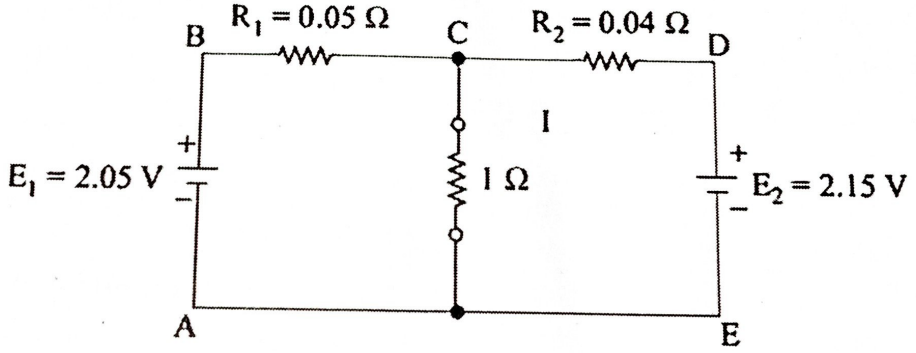
MALPRACTICE

General Instructions if any:

1. **Non-Programmable calculator is permitted: YES**
2. **Reference tables are permitted: NO**
3. **Assume any missing values**

PART – A

Answer All FIVE Questions (5 X 12 = 60 Marks)

S. No.		Questions	Marks
1	(a)	Find the Thevenin's equivalent circuit of the circuit shown in Figure 1, to the left of the terminals a and b. Also, find the current through R_L if the load resistances are 24Ω and 36Ω.  Figure 1	12
		(OR)	
	(b)	Two batteries of EMF 2.05 V and 2.15 V, having internal resistances of 0.05 Ω and 0.04 Ω, respectively, are connected together in parallel to supply a load resistance of 1 Ω as shown in Figure 2. Calculate using the superposition theorem, the current supplied by each battery, and also the load current.  Figure 2	12
2	(a)	A resistance of 40Ω, inductance of 35mH, and capacitance of 5μF are connected in series across a 220V, 50Hz AC Supply. Find (a) Impedance of circuit, (b) Power consumed in the circuit, (c) Power factor, (d) Voltage drop across resistance, Inductance, and Capacitance. Also, draw the phasor diagram for the circuit	12
		(OR)	
	(b)	At t=0, switch 1 in Figure 3 is closed, and switch 2 is closed 4 s later. Find i(t) for t>0. Calculate i for t = 3s and t = 6s	12

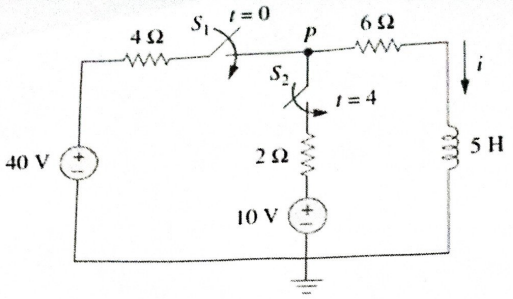


Figure 3

3	(a)	A rectangular-shaped iron core has an air gap of 0.01 cm, as shown in Figure 4. The mean length of the flux path through iron is 39.99 cm. The relative permeability of iron is 2000. The coil has 1000 turns. The cross-sectional area of the core is 9 cm^2. Calculate the current required to produce a flux of 1 mWb in the core.	8

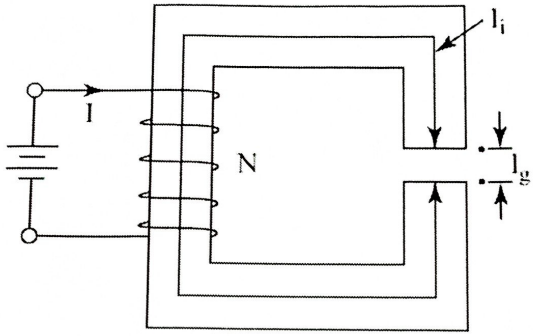


Figure 4

	(b)	Explain the working of a single-phase transformer with a neat sketch.	4
		(OR)	
	(c)	With a neat and clean diagram, explain the construction, operation, and applications of a DC motor.	12
4	(a)	Discuss the operation of a Half-Wave Rectifier and a full-wave rectifier with diagrams and graphs as required.	12
		(OR)	
	(b)	Explain the common emitter configuration in BJT, and also explain its input and output characteristics with a suitable diagram.	12
5	(a)	Draw the logic circuit for the given Boolean expression and also write the truth table. $Y = A\,C + B\,C$ Also, realize this expression with the use of either NAND gates or NOR gates only.	6
	(b)	Draw the Circuit of the S-R flip-flop using NAND gates and determine the following: truth table, excitation table, and the characteristic equation	6
		(OR)	
	(c)	Explain the Full Adder with the necessary truth table, Logic diagram, and obtain the expression for SUM and CARRY.	6

	(d)	Design 8:1 MUX using 4:1 mux and other low order MUX or logic gates.	6

PART – B

Answer ANY FIVE Questions (40 Marks)

6	Find the current flowing through the 20 Ω shown in Figure 5 using the superposition theorem	8

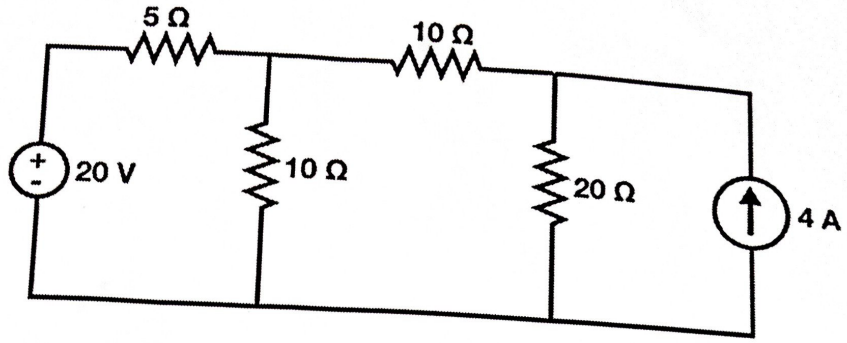


Figure 5

7	A variable resistance R and an inductance L of 100 mH are connected in series across a 50 Hz supply as shown in Figure 6. Calculate the value of R at which the voltage across the inductor is half the supply voltage.	8

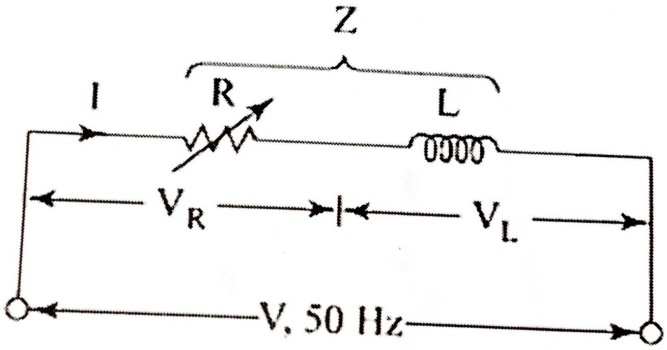


Figure 6

8	Explain the Principle of operation and working of the single-phase induction motor.	8
9	Explain the operation of a PN junction diode. How is it different from a Zener diode? Draw the V-I characteristics of a PN junction diode.	8
	(i) Convert the $(100110.01011)_2$ to its equivalent Octal Number.	
	(ii) Convert $(FED.ABE)_{16}$ to its equivalent Decimal numbers.	
	(iii) Convert $(ABCD)_{16}$ to its equivalent octal and binary numbers	
	(iv) Convert $(57.23)_8$ to its equivalent Decimal and binary numbers.	

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